

Graphene-Supported Nitrogen and Boron Rich Carbon Layer for Improved Performance of Lithium–Sulfur Batteries Due to Enhanced Chemisorption of Lithium Polysulfides

Shouyi Yuan, Junwei Lucas Bao, Lina Wang, Yongyao Xia, Donald G. Truhlar,* and Yonggang Wang*

The high theoretical energy density (2600 Wh kg⁻¹) and the low cost of sulfur have made lithium–sulfur (Li–S) batteries an attractive option to satisfy the need for advanced energy storage.^[1] However, there are several challenges. Foremost among these is the shuttling effect arising from the dissolution of lithium polysulfides intermediate into the electrolyte from the cathode reaction and their diffusion to the anode where they react with metal lithium to form lower-ordered lithium sulfides that then return to the cathode. Second is the large volume change of sulfur particles during the charge/discharge process, and third is the insulating nature of sulfur and its discharge products. Lack of a solution to these challenges has restricted the practical application of Li–S batteries because of their adverse effects on both cycle life and rate performance.^[1] Various methods have been advanced to address these issues, including the design of carbon-based host materials,^[2–4] optimization of the electrolytes,^[5] anode protection,^[6] and the design of novel battery structures.^[7] Among these approaches, porous carbon host^[2] are particularly promising because not only can they suppress the dissolution of lithium polysulfides in their large internal surface, but also they can improve the utilization of sulfur by keeping it electrically connected. However, the weak physical adsorption of the highly polar hydrophilic lithium polysulfides (i.e., Li₂S_x, 4 ≤ x ≤ 8) on the nonpolar hydrophobic carbon matrix slows down the sulfur loss only in the short term.^[4b] We especially note that very recent studies^[8,9] have revealed that the detachment of the polar lithium sulfides (i.e., Li₂S, Li₂S₂) from the nonpolar carbon hosts results in irreversible active mass loss and electrical contact isolation during

the charge process, and this is considered to be another factor contributing to the rapid decay of Li–S batteries.

Recently, it has been reported that nitrogen-doped carbon exhibits strong chemisorption for polysulfides and Li₂S, and consequently longer cycling performance has been achieved by developing N-doped carbon/S composites.^[10] It has been demonstrated that the electronegative N atoms in the carbon matrix are particularly effective in forming S_xLi⁺⋯N interactions via the lone-pair electrons of N because of the strong interaction due to Lewis acid-base interaction between Li⁺ cations in Li₂S_x and N atoms in the carbon matrix.^[10] Alternatively, boron atoms, as an electron-deficient alternative to N atoms, are electropositive in the carbon framework, thus leading to the chemisorption of polysulfide anions (S_x²⁻).^[11] Accordingly, it was reported very recently that a boron-doped carbon host can also improve the performance of S composite cathodes.^[11]

As reviewed above, recent carbon host design strategies based on the enhanced chemisorption of lithium polysulfide (Li₂S_x, 4 ≤ x ≤ 8) and Li₂S rely solely on either the chemisorption of Li⁺ cations in Li₂S and lithium polysulfides at negatively polarized N atoms or on the chemisorption of polysulfides anion at positively polarized B atoms. If N atoms and B atoms are both introduced into the carbon, the chemical adsorption for both Li⁺ cations and polysulfides anions can be increased simultaneously, which may further enhance the cycling performance of the C/S composite cathode. We especially note that if the B atoms are adjacent to the N atoms to form an N = B/N-B structure in the carbon framework, the polarization of both B atoms and N atoms will be further enhanced because the electronegativity of a N atom is higher than that of a C atom, whereas the electronegativity of a B atom is lower than that of a C atom. Therefore, it is expected that an N = B/N-B structure in a carbon framework will exhibit much higher chemisorption for both lithium polysulfides and Li₂S. Hence, we present a strategy to chemically bind sulfur and its discharge products with a graphene-supported N/B codoped carbon layer with unique N = B/N-B sites and various kinds of N sites (pyrrolic N and pyridinic N) in the surface carbon framework.

It would be a great challenge to synthesize the N = B/N-B structure by introducing the N and B separately into carbon framework because of the low percentage of heteroatoms in N/B-doped carbon materials. Therefore, our synthetic strategy is quite different. In our strategy, ionic liquid (IL, in particular [Emim]BF₄) was first introduced to functionalize a graphitic

S. Yuan, Dr. L. Wang, Prof. Y. Xia, Prof. Y. Wang
Department of Chemistry and Shanghai Key Laboratory
of Molecular Catalysis and Innovative Materials
Institute of New Energy, Collaborative Innovation
Center of Chemistry for Energy Materials
Fudan University
Shanghai 200433, China
E-mail: ygwang@fudan.edu.cn
J. L. Bao, Prof. D. G. Truhlar
Department of Chemistry
Chemical Theory Center, and Supercomputing Institute
University of Minnesota
State of Minnesota, MS 55455-0431, USA
E-mail: truhlar@umn.edu



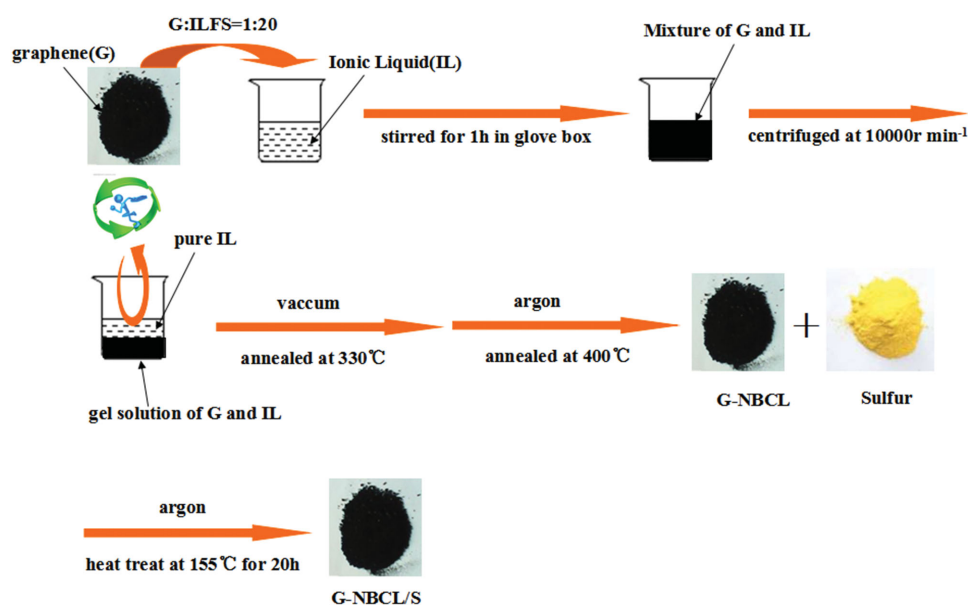
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surface to form an IL/graphene composite, and this was then converted into a graphene-supported N/B rich carbon layer (G-NBCL) by a two-step annealing procedure. Subsequently, the G-NBCL was employed as a sulfur host to prepare the C/S composite cathode with superior electrochemical performance. While the G-NBCL composites exhibit the high conductivity of the graphene template, the surface of G-NBCL is rich in nitrogen (pyrrolic N, pyridinic N, and N = B/N-B species) and boron heteroatoms that are confirmed to have strong chemical binding to lithium polysulfides (Li_2S_x , $4 \leq x \leq 8$) and lithium sulfides (i.e., Li_2S , Li_2S_2) by both quantum mechanical calculations and visual discrimination. In addition, the quantum mechanical calculations further demonstrate that the unique N = B/N-B structure in the G-NBCL exhibits much higher binding energy for both lithium polysulfides and lithium sulfides than that of pyrrolic N, pyridinic N, and pristine graphene. As a result, the as-prepared G-NBCL/S composites with a sulfur content of 70 wt% and a reasonable loading based on the weight of sulfur in the cathode of $0.9\text{--}1.2\text{ mg cm}^{-2}$ display an ultralong cycling performance of over 1500 cycles with a low decay rate of 0.035% per cycle at the rate of 2 C and a high rate performance up to 3 C even at this pre-optimization stages (i.e., no special binder, no special electrolyte additive, etc.)

The G-NBCL composites were experimentally realized via a two-step annealing procedure of ionic liquids ([Emim]BF₄) on the graphene surface as schematically illustrated in Scheme 1. As shown in Scheme 1, the graphene, prepared by thermal treatment at 800 °C in argon for 2 h, was first dispersed in the ionic liquids ([Emim]BF₄) to form a suspension. After centrifuging at 10 000 r min⁻¹, a transparent pure ionic liquid was obtained in the upper layer, which can be recycled for the next time, while the black gel phase in the lower layer, identified as the mixture of ionic liquid and graphene, was heat treated at 330 °C under vacuum for 50 min and then further annealed at 400 °C for 1 h in argon to form a graphene-supported N/B rich carbon layer (G-NBCL400). To investigate the effect of annealing temperature

on the morphology, structure, and the electrochemical performance, the G-NBCL with final heat-treat temperature of 900 °C (G-NBCL 900) was also prepared with the same method for comparison. Previous work^[12] showed that the static-assistant CH-p interaction between the ionic liquid and the highly graphitic surface enables the entrapping and redirecting of the ionic liquid and modifies the thermal behavior of ionic liquid. In the G-NBCL composites, the highly graphitic graphene surface serves as a template, while the ionic liquid ([Emim]BF₄) acts as a source of heteroatom and surface carbon precursors. The graphene in the G-NBCL has two functions: first, the graphitic layers in the graphene serves as a reservoir to entrap the ionic liquid. Second, the high conductive graphene basement also improves the conductivity of the newly form N,B codoped carbon layers. The as-prepared G-NBCL was then ground with sublimed sulfur in a ratio of 3:7 with a mortar, and then heat-treated at 155 °C for 20h to fully contact the sulfur with G-NBCL (i.e., G-NBCL400/S and G-NBCL900/S composites). As the G-NBCL is highly sensitive to water, all the preparing procedures were carried out in a glove box filled with pure argon as far as possible. For comparison, the nitrogen doped graphene (NDG) and boron doped graphene (BDG) were also synthesized according to the previous report^[11] by using the urea as nitrogen source and boron acid as boron source, respectively. The NDG/S and BDG/S composites were also synthesized by the same method as G-NBCL/S composites. The sulfur content in the G-NBCL/S composite was determined to be 70 wt% by thermogravimetric analysis (TGA) (see Figure S1, Supporting Information).

Transmission electron microscopy (TEM) was conducted to analyze the morphology of the G-NBCL400 and G-NBCL400/S composites. As shown in Figure 1a, the structure of G-NBCL400 is a 2D layered and crumbled nanosheet with a few layers stacked, which was further confirmed by SEM images (Figure S2, Supporting Information). Notably, some nanoparticles were observed on the nanosheet in the TEM image



Scheme 1. Preparation procedures of G-NBCL/S composites.

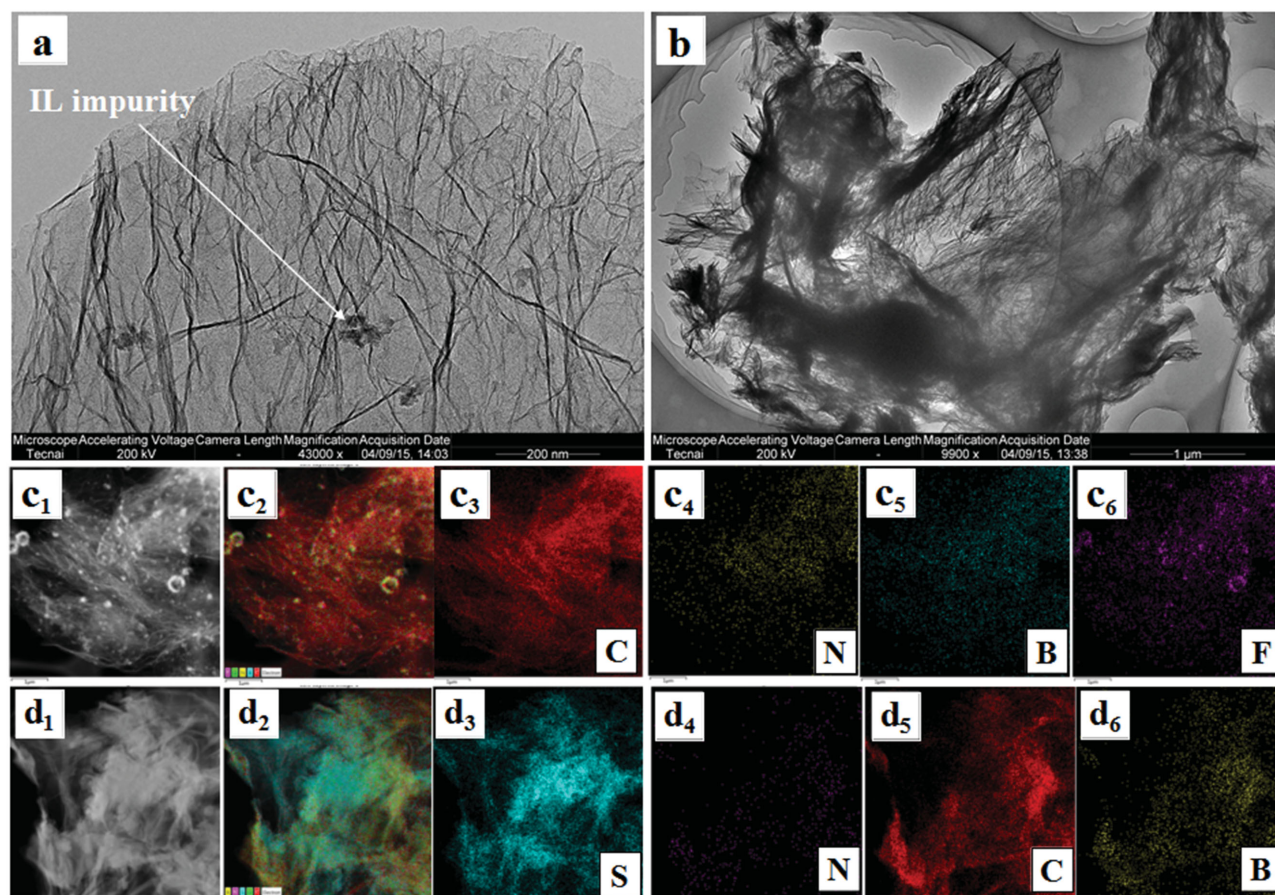


Figure 1. TEM images of G-NBCL and G-NBCL/S and EDS mapping. a) TEM image of G-NBCL composites. b) TEM image of G-NBCL/S composites. c₁–c₆) EDS mapping of G-NBCL composites; c₁) EDS image, c₂) EDS mapping of C, N, B, F, c₃) EDS mapping of C, c₄) EDS mapping of N, c₅) EDS mapping of B, c₆) EDS mapping of F. d₁–d₆) EDS mapping of G-NBCL/S composites; d₁) EDS image, d₂) EDS mapping of C, N, B, S, d₃) EDS mapping of S, d₄) EDS mapping of N, d₅) EDS mapping of C, d₆) EDS mapping of B.

(Figure 1a); this is attributed to ionic liquid impurities that were not fully decomposed and polymerized during the process of thermal treatment.

To further investigate the components of the G-NBCL400 composite, energy dispersive X-ray spectrum (EDS) mapping was carried out. As shown in Figure 1c₁–c₆, the nanosheet contains various elements including nitrogen, oxygen, boron, and fluorine heteroatoms, and nitrogen, oxygen and boron elements are uniformly distributed on the nanosheets, while fluorine is mainly in the small particles on the surface arising from the ionic liquid impurity. Figure S3 of the Supporting Information shows high-resolution transmission electron microscope (HRTEM) images of both graphene and the G-NBCL400 composites. These images show that the pristine graphene (Figure S3a,b, Supporting Information) is composed of a highly graphitic surface with a *d*-spacing of ≈ 0.5 nm after annealing at 800 °C in argon. However, in the G-NBCL400 (Figure S3c,d, Supporting Information), some amorphous layers can be detected on the edge of the graphene, and these are interpreted as the polymerization and carbonization of the ionic liquid on the edge of graphene after thermal treatment at 400 °C. At the same time, the graphitic layers of graphene can be still observed in the G-NBCL400, and these can facilitate electron transfer from

and to the poorly conductive S. In order to probe the effect of annealing temperature on the morphology of G-NBCL composites, the morphology of G-NBCL900 composites were also investigated (see Figure S4 and corresponding discussion in the Supporting Information). For comparison, the morphology of NDG and BDG were also investigated (see Figures S5 and S6 and corresponding discussion in the Supporting Information for details).

Figure 1b,d₁–d₆ displays the TEM image and EDS mapping of G-NBCL400/S composites. Both the TEM image and EDS mapping indicate that the sulfur was distributed in the matrix of G-NBCL400, which is further confirmed by the SEM images of G-NBCL400/S (Figure S7, Supporting Information). Notably, even after strong sonicating in ethanol, TEM showed that the sulfur particles were still anchored in the matrix of G-NBCL400, indicating the strong adhesion of sulfur on the G-NBCL400.

To further investigate the surface component of G-NBCL400, X-ray photoelectron spectroscopy (XPS) was carried out. Figure 2a–d shows the B 1s, N 1s, and C 1s XPS spectrum of G-NBCL400 and S 2p XPS spectrum of G-NBCL400/S composites. The B 1s XPS spectrum in Figure 2a displays a broad peak ranging from 187 to 196 eV, which can be fitted as a sum of three peaks assigned to N = B/N-B (≈ 190.8 eV), N-B (≈ 191.8 eV),

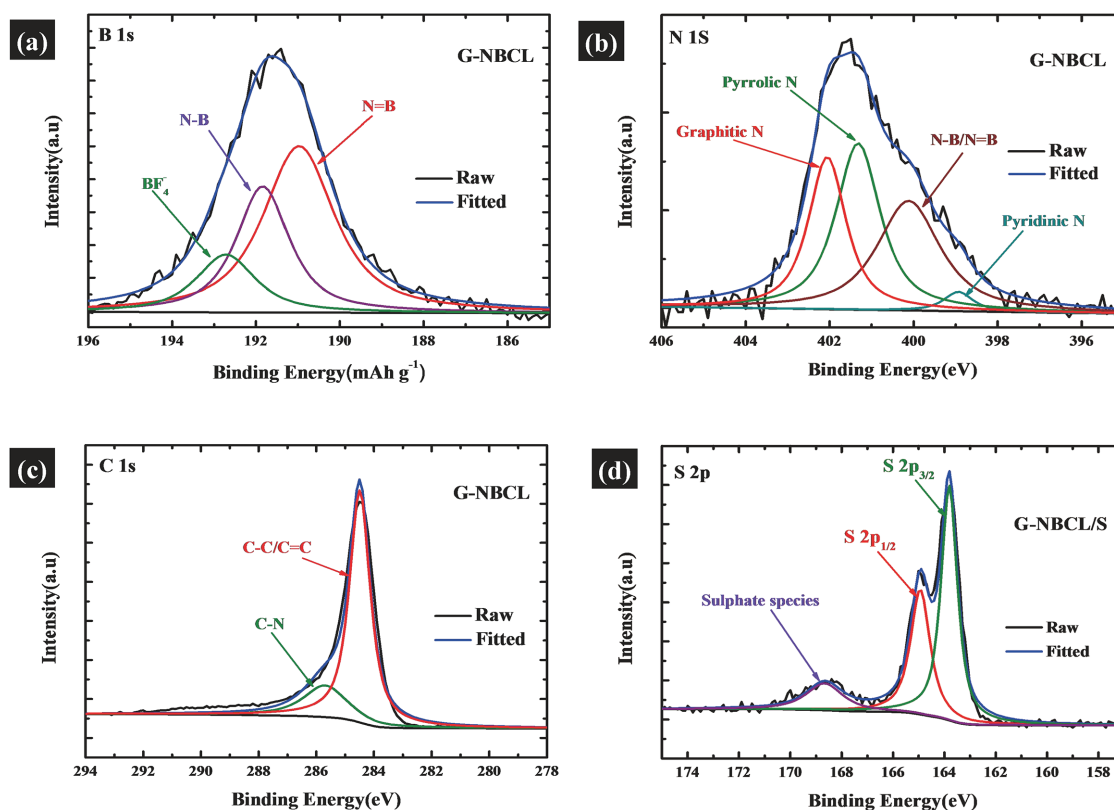


Figure 2. XPS investigation of G-NBCL and G-NBCL400/S. a) N 1s XPS spectrum of G-NBCL400 composites. b) B 1s XPS spectrum of G-NBCL400 composites. c) C 1s XPS spectrum of G-NBCL400 composites. d) S 2p XPS spectrum of G-NBCL400/S composites.

and BF_4^- (193.0 eV), respectively.^[12] The N 1s XPS spectrum in Figure 2b also shows a broad peak, this one ranging from 396 to 406 eV. It can be fitted by four peaks assigned to pyridinic N (≈ 398.6 eV), N = B/N-B (≈ 400.0 eV), pyrrolic N (≈ 401.1 eV), and graphitic N (≈ 402.1 eV), respectively.^[12] These N and B functional groups can chemically bind the sulfur and its discharge products in the carbon framework of G-NBCL400 composites. The C 1s spectrum of G-NBCL400 in Figure 2c can be fitted as a sum of two peaks, in particular one at ≈ 285.8 eV assigned to the C–N bond and one at ≈ 284.5 eV assigned to conjugated C–C bonds (C–C/C=C) of sp^2 hybridized carbon. Notably, no peaks corresponding to C–O bonds (i.e., ≈ 287.2 eV for C=O, ≈ 286.1 eV for C–O, ≈ 288.6 eV for O–C=O)^[3d] can be observed in the C 1s spectrum, suggesting the successful removal of oxygen-containing groups on the graphene oxides by the thermal treatment at high temperature. However, C 1s XPS spectrum of the G-NBCL400/S composites (Figure S8b, Supporting Information), the peak at ≈ 285.8 eV in the G-NBCL400/S, is much stronger than the corresponding peak in the G-NBCL400 composites, which arises in that case from the overlapped peaks of C–N and C–S bonds. The presence of C–S bond will bind the sulfur in the carbon frameworks. According to the XPS results and previous work,^[12] N, B, and C atoms are totally exposed on the surface of G-NBCL400 and occur in the same hexagonal ring. As calculated from the XPS peak area and the sensitivity factors (which are 0.25 for C, 0.13 for B, 0.42 for N, and 0.66 for O), the atomic percentages of C, N, B, and O in the G-NBCL400 composites are 87.4%,

5.3%, 2.9%, and 4.5%, respectively. Figure 2d shows the S 2p spectrum of G-NBCL400/S composites. This spectrum shows two peaks with a separation of 1.3 eV; these peaks are assigned to S $2p_{1/2}$ and S $2p_{3/2}$, respectively. It can be observed that the binding energy of S $2p_{1/2}$ in the G-NBCL400/S is 163.7 eV, which slightly shifts to the lower binding energy compared with the corresponding peak for elemental S (164.0 eV); this indicates the possible presence of C–S bonds.^[8b] The peak from 167 to 171 eV is assigned to sulphate species arising from sulfur oxidation in the air.^[8b] The N 1s, B 1s, and C 1s XPS spectra of G-NBCL400/S and the C 1s XPS spectrum of graphene/S (G/S) are given in the Supporting Information (see Figure S8 and corresponding discussion in the Supporting Information). In order to investigate the effect of annealing temperature on the structure of G-NBCL composites, the G-NBCL900 composites were also characterized by the XPS spectrum (see Figure S9 and corresponding discussions in the Supporting Information). To further probe the structure of G-NBCL400 and G-NBCL400/S composites, the Fourier transform infrared (FT-IR) spectrum and the X-ray diffractions (XRD) pattern were also investigated, and the Brunauer–Emmett–Teller (BET) area and pore distribution were also measured (see Figures S10 and S11 and corresponding discussion in the Supporting Information). Finally, the Raman spectrum was also carried out on the pristine G, G-NBCL400, and G-NBCL900 composites (see Figure S12 and corresponding discussion in the Supporting Information).

The as-prepared G-NBCL400/S composites were then served as a cathodic materials for Li–S batteries. 2016 coin

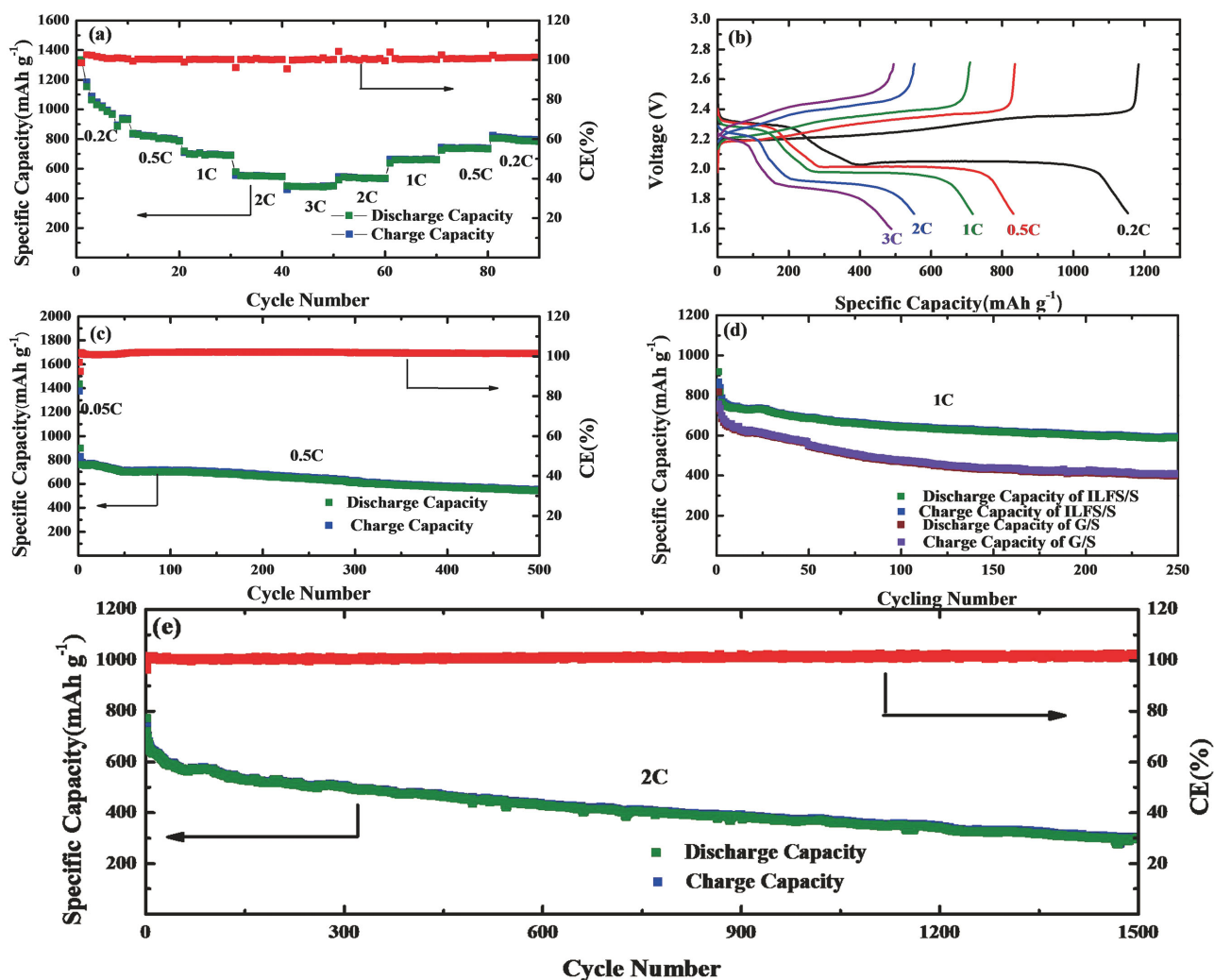


Figure 3. Electrochemical performance of G-NBCL400/S composites. a) Galvanostatic cycling performance of G-NBCL400/S composites at various rates ranging from 0.2 to 3 C. b) Voltage profiles of G-NBCL400/S composites at various rates. c) Cycling performance of G-NBCL400/S composites at a rate of 0.5 C. d) Comparison of the cycling performances of G-NBCL400/S composites and G/S composites at a rate of 1 C. e) Cycling performance at a rate of 2 C.

cells were fabricated with a typical sulfur content of 70 wt% and a reasonable sulfur loading of 0.9–1.2 mg cm⁻². Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) (1 M) dissolving in 1,2-dimethoxyethane (DME) and 1,3-dioxolane (DOL) (1:1) with 2 wt% lithium nitrate (LiNO₃) was employed as the electrolyte. The role of LiNO₃ is to help reduce the shuttle effect and passivate the lithium metal anode.^[5a] **Figure 3a** shows the rate performance of the G-NBCL400/S composites ranging from 0.2 to 3 C (1 C = 1672 mAh g⁻¹). The cell delivers a capacity of 1050 mAh g⁻¹ at the rate of 0.2 C, and then slowly decreases to 833 mAh g⁻¹ at the rate of 0.5 C, to 699 mAh g⁻¹ at the rate 1 C, and to 555 mAh g⁻¹ at the rate of 2 C. Even at high rate of 3 C, a reversible capacity of 480 mAh g⁻¹ can be still achieved. Moreover, when the rate restored to 0.2 C, the G-NBCL400/S composite cathode almost completely recovers its original capacity. It can be noted in **Figure 3a** that the sulfur cathode still suffers from a rapid decay in the first several cycles. The fast capacity fading in the first 10–20 cycles mainly arises

from the dissolution of sulfur and polysulfides into the electrolyte over first several cycles, which will then suppress the further dissolution and diffusion of polysulfide over following cycles. As reported previously,^[10a] the polysulfides can also be employed as the electrolyte additive to eliminate this capacity decay in the first cycles. **Figure 3b** displays the corresponding voltage profiles of G-NBCL400/S composites at different rates ranging from 0.2 to 3 C; two typical plateaus of S cathode, associated with the reduction from the cyclic S₈ to the long chain lithium polysulfides and from the long chain lithium polysulfides to the short chain lithium sulfide, can be observed in the voltage profiles at all current rates. We see that increasing the rate decreases both the discharge capacity and the discharge plateau; these results can be explained by the larger over-potential and the lower S utilization at the higher current rate. However, even at the high rate of 3 C, a reversible capacity of 480 mAh g⁻¹ can still be achieved, which is almost the best performance ever reported.^[2a,c-f,3a-c,e,4,5,7b-d,11] The excellent rate performance

is attributed to both the enhancement of conductivity by the high conductive graphene and the strong affinity between the lithium sulfide and the G-NBCL400.

For the practical application of Li-S batteries, good capacity retention over long cycles is essential. Figure 3c shows the performance of G-NBCL400/S composite cathodes over 500 cycles at the rate of 0.5 C. The cell was first activated at 0.05 C, and then it was cycled over 500 times at 0.5 C. After activating at 0.05 C, an initial discharge capacity of 829 mAh g⁻¹ is achieved at 0.5 C, which then stabilizes at 760 mAh g⁻¹ after five cycles. The capacity then slowly decreased to 627 mAh g⁻¹ after 300 cycles, corresponding to a capacity retention of 82%. Even after 500 cycles, a capacity of 556 mAh g⁻¹ is still achieved, representing an extremely low decay rate of 0.053% per cycle. Figure 3d exhibits the cycling performance of both G-NBCL400/S composites and graphene/S (G/S) composites at the rate of 1 C. This figure shows that both the G-NBCL400/S composite and the G/S composite display similar initial discharge capacity. However, after 250 cycles at the rate of 1 C, a specific discharge capacity of 588 mAh g⁻¹ is still available for G-NBCL400/S

composites, corresponding to a capacity retention of 80%. However, for G/S composites prepared by the same procedure, only a specific capacity of 398 mAh g⁻¹ remains after 250 cycles. The rapid capacity loss in the G/S composites should be attributed to both the weak physical adsorption of lithium polysulfides on graphene and the detachment of the discharge products from the nonpolar carbon matrix. When the rate is increased to 2 C (Figure 3e), performance was maintained over 1500 cycles. The battery displays an initial discharge capacity of 744 mAh g⁻¹, which then stabilizes at 650 mAh g⁻¹ after 5 cycles. Even after 1500 cycles, a specific capacity of 301 mAh g⁻¹ is still available, representing a low capacity decay of 0.035% per cycles. (The decay rate was calculated from five cycles after achieving a stabilized capacity.) The long-term cycling stability is ascribed to both the strong chemisorption of the polysulfides on G-NBCL400 and the strong affinity between the G-NBCL400 and lithium sulfide (i.e., Li₂S, Li₂S₂). The cycling performance at the rate of 0.2 C and the voltage profiles over 500 cycles at the rate of 0.5 C are given in Figure S13 of the Supporting Information. Although 0.035% is not the lowest decay rate ever reported, the

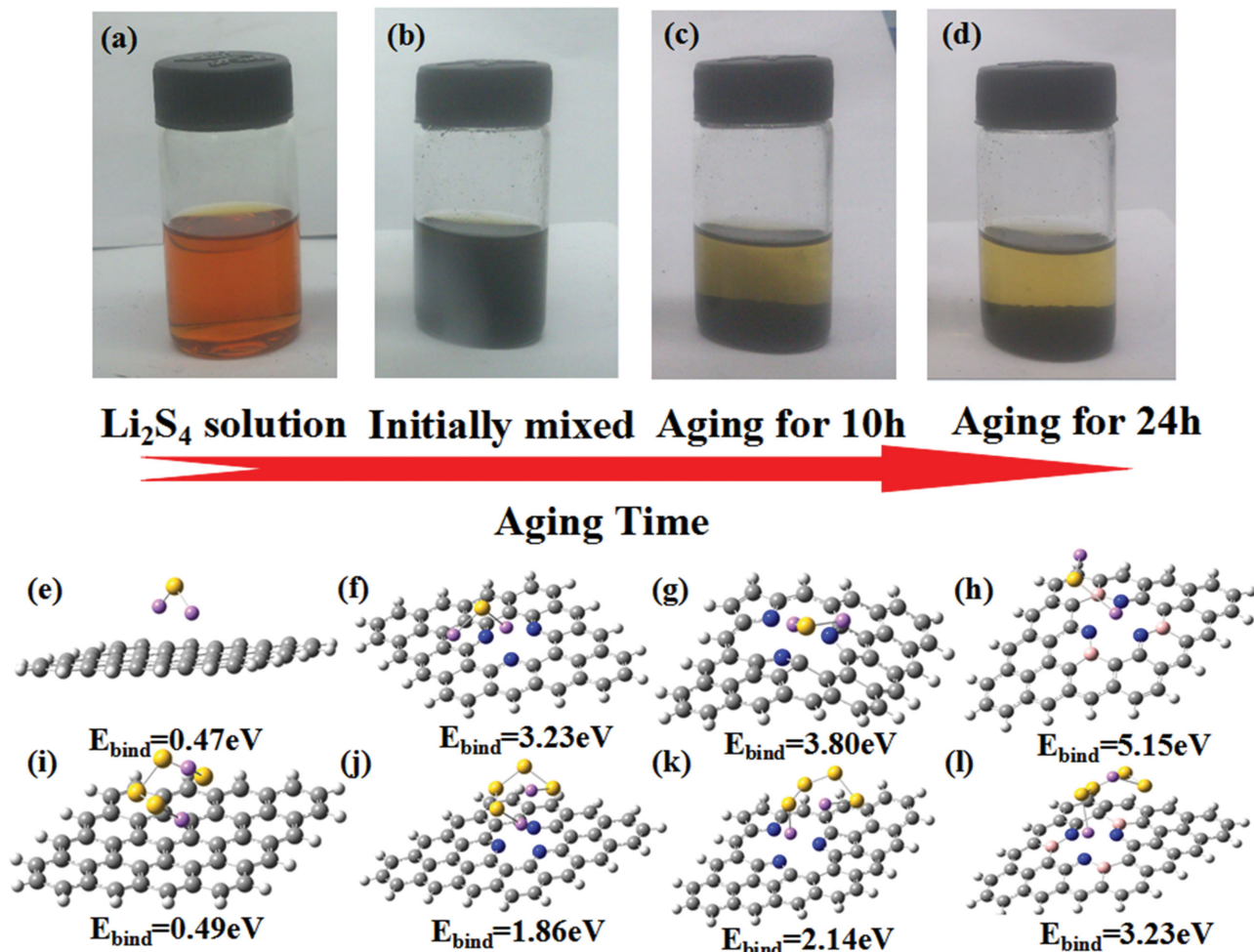


Figure 4. Visual discriminations and quantum mechanical electronic structure calculations. a–d) Visual discrimination of Li₂S₄ solution in THF, both initially and at various times after adding G-NBCL400. e) Binding energy of Li₂S on pristine graphene; f) Binding energy of Li₂S at a pyridinic N site; g) Binding energy of Li₂S at a pyrrolic N site; h) Binding energy of Li₂S at an N = B/N-B site; i) Binding energy of Li₂S₄ on pristine graphene; j) Binding energy of Li₂S₄ at a pyridinic N site; k) Binding energy of Li₂S₄ at a pyrrolic N site; l) Binding energy of Li₂S₄ at an N = B/N-B site.

performance of G-NBCL400/S composites with a reasonable sulfur loading is among the highest performance ever reported^[3a-c,e,3a-c,e,4b,c,5,7,8b,9,11] even at this preoptimization stage of development. Hence, comparison of the decay rate is difficult in the light of different current rates, sulfur loading in the electrode, sulfur content in the composites, battery configuration and electrolyte. For comparison, the NDG/S composites and BDG/S composites were also cycled at 0.5 C and the electrochemical performance is given in the Supporting Information. It can be identified that the cycling performances of NDG/S and BDG/S composites were inferior to that of G-NBCL400 composites, indicating that the binding energy of G-NBCL for polysulfides is much higher than that of NDG and BDG (see Figure S14 and corresponding discussion in the Supporting Information for details). The cycling performance of G-NBCL400 with high sulfur loading is also investigated. The loading based on the weight of sulfur is 1.7 mg cm^{-2} , corresponding to a total mass loading of 3.0 mg cm^{-2} (see Figure S15 and corresponding discussion in the Supporting Information for details). In order to further investigate the effect of annealing temperature on the electrochemical performance, the electrochemical performance of G-NBCL900/S composite was also investigated. The achieved results indicate G-NBCL900/S exhibits the cycle performance that is close to that of G-NBCL400/S (see Figure S16 and corresponding discussion in the Supporting Information). Compared with previous reports concerning N doped carbon/S or boron doped Carbon/S composites (Table S1, Supporting Information), the G-NBCL400/S shows long cycle stability and high rate performance. It can be identified that the sulfur utilization of G-NBCL composites is slightly lower than other works,^[10] which is due to the large particles size of sulfur in the G-NBCL/S composites confirmed by SEM images of G-NBCL/S composites (Figure S7, Supporting Information). Therefore, to further improve sulfur utilization, more investigations should focus on the synthesis of nanosized the sulfur particles.

The superior cycling performance of the cells is mainly attributed to the strong polysulfide adsorption on G-NBCL400 and the strong affinity between G-NBCL400 and Li_2S . To confirm the strong adsorbability of G-NBCL400 for lithium polysulfides, both visual discrimination and quantum mechanical calculations were carried out.

To experimentally illustrate the strong adsorption for polysulfides, a Li_2S_4 solution, prepared by stoichiometrically dissolving Li_2S and sublimed S in a molar ratio of 1:3 in anhydrous tetrahydrofuran (THF), was taken as a representative of lithium polysulfides. Then, 20 mg G-NBCL was added into 10 mL Li_2S_4 solution. Figure 4a–d displays color of pristine Li_2S_4 solution and the color changes of the Li_2S_4 solution at various periods after adding G-NBCL composite. After adding G-NBCL into the Li_2S_4 solution, a homogeneous black suspension was initially formed, and it took several minutes for G-NBCL to precipitate. After aging for 10 and 24 h, the color of Li_2S_4 turns from the initial orange to light yellow, indicating that much of the Li_2S_4 has been adsorbed by the G-NBCL. The result indicates the strong adsorption for lithium polysulfides. We attribute this to a synergy of factors: First, the unique 2D structure of the G-NBCL composite with abundant pores, as confirmed by both TEM images and BET measurements, enables the physical adsorption of the Li_2S_4 in the pore of

G-NBCL. Second, B atoms in the G-NBCL have a lower electronegativity than either N or C, and they are positively polarized, leading to chemisorption of negative species on the surface of G-NBCL. As S and polysulfides anions are both electron-abundant,^[12] they can be adsorbed by the positive boron sites. Finally, the pyrrolic N and pyridinic N functional groups can chemically adsorb the Li^+ cations in the lithium polysulfides. To further confirm the strong binding capability of G-NBCL for polysulfides, ex situ XPS was also carried out (see Figure S17 and corresponding discussion in the Supporting Information for details).

To further confirm and quantitatively characterize the strong chemical adsorption of the discharge product Li_2S and the intermediate lithium polysulfides on G-NBCL, quantum mechanical electronic structure calculations were carried out with Kohn–Sham density functional theory using the Gaussian 09 software package.^[13] The global-hybrid meta-GGA exchange-correlation functional M06-2X,^[14] which has excellent performance for describing the noncovalent interactions in a wide range of systems,^[15] was chosen to calculate the binding energy of the representative lithium polysulfides Li_2S_4 and Li_2S on G-NBCL. The counterpoise method^[16] of Boys was employed to correct for basis set superposition error. (Binding energies calculated without this correction are given in the Supporting Information (Table S2), and they do not change the conclusions.) In addition, the Cartesian coordinates of optimized geometries at M06-2X/6-31+G(d,p) level are also given in Table S3 of the Supporting Information. As demonstrated by the XPS investigation (Figure 2), there are three kinds of N sites in the G-NBCL, namely, pyridinic N sites, pyrrolic N sites, and N = B/N-B sites.

The formation mechanism of the unique N = B/N-B structure has been described in detail previously,^[12] and based on our XPS results and on the previous work,^[12] we know that the B, N, and C atoms are present in the same hexagonal ring in the N = B/N-B structure. At the same time, although the newly formed heteroatom-rich carbon layer is abundant in defect, the graphitic fragments on the carbon layer are continuous. Figure 4e–h displays the molecular geometries and the binding energies of Li_2S on pristine graphene (Figure 4e), at a pyridinic N sites (Figure 4f), at a pyrrolic N sites (Figure 4g), and at an N = B/N-B sites (Figure 4h). The weak binding energy of Li_2S on graphene surfaces results in the detachment and separation of electroactive Li_2S from nonpolar conductive undoped carbon matrices, and this results in incomplete conversion from Li_2S to S during the charging process, which greatly decreases the cycle life of Li–S batteries. Contrarily, the strong adhesion between G-NBCL and Li_2S will keep Li_2S in good electrical contact with the conductive G-NBCL during the charge. Similarly, Figure 4i–l shows the molecular geometry and the binding energy Li_2S_4 binding to pristine graphene (Figure 4i), a pyridinic N sites (Figure 4j), a pyrrolic N sites (Figure 4k), and an N = B/N-B sites (Figure 4l). The Cartesian coordinates of the optimized geometries are given in the Supporting Information. As shown in Figure 4e–f, the binding energy of Li_2S on graphene, which is 0.47 eV, is much lower than that for binding to the G-NBCL400, which is 3.23 eV for pyridinic N sites, 3.80 eV for pyrrolic N sites, and 5.15 eV for the N = B/N-B sites. On the other hand, as shown in Figure 4i–l, the binding energy of Li_2S_4 on the G-NBCL (1.86 eV at pyridinic N sites, 2.14 eV at pyrrolic N sites, and 3.23 eV at N = B/N-B sites) is also much higher

than that of graphene (0.49 eV), suggesting the strong chemisorption of lithium polysulfides on the G-NBCL.

The quantum mechanical calculations further conform the strong binding of the discharge products to the G-NBCL carbon host. Furthermore, they show that the unique N = B/N-B sites in the G-NBCL exhibits the strongest binding affinity for both lithium polysulfides and Li₂S. Therefore, we conclude that the strong binding energy of both lithium polysulfides and lithium sulfides on G-NBCL is mainly responsible for the superior electrochemical performance of G-NBCL400/S composites.

In conclusion, we have demonstrated a strategy to chemically bind a sulfur cathode and its discharge products with a graphene-supported N/B rich carbon layer. Compared with previous reports (Table S1, Supporting Information), the cathode designed this way shows a high rate and long cycle performance. The superior performance should be attributed to a synergy of factors including the high conductivity of the graphene template, rich wrinkles of the G-NBCL to accommodate the large volume change and the strong affinity of G-NBCL for the discharge products, which is confirmed by both visual discrimination and quantum mechanical calculations. This opens a new path to a departure from traditional cathode design for Li-S battery based on the monodoped carbon hosts (primary N doped carbon host) and should inspire further studies on the nature of the multi-doped carbon host for Li-S battery.

Experimental Section

Synthesis of Graphene Oxides: Graphene oxides were synthesized by Hummer's method, improved as explained previously.^[17] Detail synthesis procedures for the graphene oxides are given in the Supporting Information.

Synthesis of Graphene: The graphitic surface was obtained from graphene oxides by heat treatment of graphene oxides at 800 °C for 2 h in argon to form a highly graphitic surface.

Synthesis of G-NBCL, BDG, and NDG: For the synthesis of G-NBCL, the as-obtained highly graphitic surface was initially mixed with 1-ethyl-3-methylimidazolium tetrafluoroborate ([Emim]BF₄) in a ratio of 1:20 in a glove box filled with argon. Then, the mixture was stirred for 30 min to get a homogeneous gel solution. The solution was then centrifuged at 10 000 r min⁻¹. A transparent liquid phase in the upper phase, which was considered to be pure ionic liquid, was recycled from the lower gel phase identified as the mixture of ionic liquid and graphene. Subsequently, the lower gel solution was put into a quartz boat and then transferred into the furnace for heat treatment at 330 °C under vacuum and then further annealed at 400 °C in argon to form the G-NBCL400 composites. In order to compare the effect of annealing temperature on the morphology of the composites, the composites was annealed at 900 °C after the heat treatment at 330 °C under vacuum to form the G-NBCL900 composites. For comparison, Nitrogen doped graphene and boron doped graphene were also prepared according to the previous report^[11] using urea and boron acid as nitrogen source and boron source, respectively.

Synthesis of G-NBCL/S, NDG/S, and BDG/S: The G-NBCL/S was obtained by a melt-diffusion strategy. Briefly, the G-NBCL was mixed with sublimed sulfur in a ratio of 3:7, and ground in mortar for 1 h in a glovebox. Then, the mixture was transferred into a sealed vessel filled with argon and heated at 155° for 20 h. For comparison, graphene/S composites were also prepared by using the same procedures. NDG/S and BDG/S were also prepared by the same procedures as G-NBCL/S composites.

Fabrication of Cathode: The as-prepared G-NBCL/S composites were mixed with Super P and polyvinylidene fluoride (PVdF) binder in a ratio of 8:1:1. Then the mixture was slurried on a carbon coated Al foil. The electrode was then dried at 50° under vacuum over 24 h to remove

the residual solvent. The average sulfur loading in the electrodes is approximately 0.9–1.2 mg cm⁻². Then, CR2016-type coin cells were fabricated in a glove box filled with pure argon; lithium foil was used as the counterpart electrode. These two electrodes were separated by a Celgard separator dipping with DOL/DME(1:1)-(1 M) LiTFSI electrolyte containing 2 wt% LiNO₃ additives. The cells were cycled on a LAND cyler Wuhan Land Electronic Co. Ltd. Galvanostatic charge–discharge measurements were made with a voltage ranging from 1.7 to 2.7 V. Specially, the voltage of cells tested at the rate of 3 C ranges from 1.6 to 2.7 V. For comparison, the batteries with NDG/S composites and BDG/S composites were also prepared by the same procedures.

Preparation of Li₂S_x Solutions: The Li₂S_x solution was prepared by stoichiometrically dissolving sublimed S and Li₂S in a molar ratio of 3:1 in an anhydrous THF solvent and set aside for several days to fully complete the reaction.

Characterization: The surface morphology of the above composites was characterized by SEM (FE-SEM S-4800) and TEM (JEOL JEM-2100 F microscope (Japan) operated at 200 kV). Elemental mapping was performed using EDS resulting in SEM and TEM. XPS measurements were carried out on a XSAM800 Ultra spectrometer.

Computational Method: The global-hybrid meta-GGA density functional M06-2X^[14,15] was employed to calculate the binding energy; the basis set for geometry optimization was 6-31+G(d,p), and the basis set for single-point energy calculations was MG3S.^[18] The binding energy is calculated using the equation

$$E_{\text{bind}} = E_{\text{A}} + E_{\text{B}} - E_{\text{AB}}$$

where *E* is the Born–Oppenheimer electronic energy (including nuclear repulsion), A and B are monomers, and AB is the dimer. Geometries of all the molecules were optimized by M06-2X/6-31+G(d,p). Calculations were made both with and without using the counterpoise correction of Boys to correct for basis set superposition error. Ultrafine integration grids are used in all the calculations. All of the quantum mechanical calculations were carried out with Gaussian 09.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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